

Circumgalactic dust halos update #2

Jacqueline McCleary

October 24, 2024

1 Executive Summary

1. Literature review – very recent developments in modeling
2. Accompanying modeling updates (and code improvements)
3. Updated results based on new models – including fitting power laws! The TL;DR is that while the *slope* for all redMaGiC-based XXX samples XXX is a very close match to the Ménard result, the *amplitude* is about twice as high. This is something we can and should look into in future work.
4. For the first time, I have also expand our search for extended dust halos to the Northern Sky! I tried a few different SDSS background samples as well as two different foreground samples: WISExSCOS and GWSLC (itself divided by specific star formation rate). foreground samples. While the WISExSCOS x SDSS results tended to be consistent with the redMaGiC samples at *low* impact parameters, at *high* impact parameters ($> XX$ arcminutes), the extinction flattens out and doesn't decrease. Meanwhile, the GWSLC x SDSS extinction has an interesting dip at intermediate impact parameters but agrees with the redMaGiC curves at high and low impact parameters.
5. Also, my talk at the Aspen Center for Physics dust workshop went really well! There were good questions and a lot of interest afterwards.

2 Literature Review

Since I prepared my last update, several exciting and highly germane papers have been published. Results include an interesting hydrosim study of dust survival in the IGM, a new model for dust extinction, and an update to the canonical SFD map.

2.1 Gordon et al. 2023

described in [2]. Implemented with the `dust_extinction` software package [1].

2.2 Helena outflow paper

2.3 CSFD

- I saw this person give a talk on the topic at the Aspen dust workshop, and then Nao Suzuki called my attention to it again at the COSMOS-Web meeting.

- The basic premise of the paper is that the classic Schlegel, Finkbeiner, & Davis (1998, SFD) dust map has significant contamination from the cosmic infrared background, that is, the two-halo term of large-scale structure.
- Apparently, this fact has been discovered and rediscovered in the literature [[10.1093/pasj/59.1.205](#), [10.1093/pasj/65.2.43](#)], including by our friend Brice Ménard [[2019ApJ...870..120C](#)].

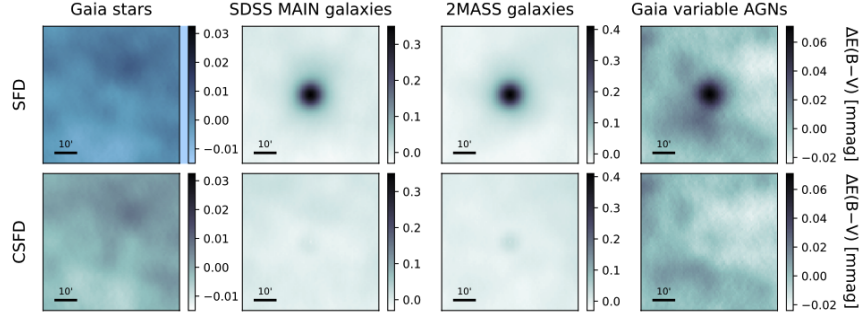


Figure 1: Stacked images of the SFD $E(B - V)$ (top row) and the CSFD equivalent (bottom row) centered on different types of sources. From left to right, the stacks are centered on Gaia stars, SDSS main sample galaxies, 2MASS galaxies, and Gaia AGNs, respectively. Aside from the stars, the SFD $E(B - V)$ values around all galaxy samples are biased high due to the LSS contamination. This bias is significantly reduced in CSFD. (Caption and figure adapted from [[2023ApJ...958..118C](#)].)

2.4 Dust as a supernova systematic

3 Code and Modeling Updates

3.1 Code updates

3.1.1 Color-matched randoms

- describe algorithm, show nice figure
- This didn't significantly affect the result, however.

3.1.2 Random catalog generation overhaul

- Mention bug of picking healpixels; could even show the bad ra/good ra figures
- Also implemented ability to generate catalogs in galactic coordinates, which made sense for the WISExSCOS catalog
- Based on the redMaGiC ratio of random galaxies to real galaxies, the random catalogs I create are $\sim 20\times$ the length of their respective data catalogs. For very large galaxy catalogs like SDSS, generating an equivalent random catalog with almost a billion objects is too much for my HPC to handle. (Or, perhaps my code is inefficient.) So, I added a little option to create multiple smaller catalogs and concatenated them.

3.1.3 Treecorr variance and misc. utilities

- Fixed variance! Used patches and compensated estimates

- `RADec_to_healpixel`
- deredden magnitudes, the script used to incorporate CSFD extinction values into magnitudes.

3.2 Model improvements

3.2.1 Gordon 2023 extinction

- At the workshop, Daniella Calzetti told me that she was flattered but that I needed to update the extinction law that I was using. She pointed me to the Gordon 2023 paper referenced above.
- I found the `dust_extinction` repository¹ and used that.
- It did shift our values, but the changes are irrelevant since the previous values were so out of date!

3.2.2 CSFD

- As mentioned earlier, Nao pointed me to this CSFD paper, where the amplitude of the residual seemed non-negligible (10^4) and so worth implementing.
- I located a nice software package called `dustmapper`² to query the CSFD dust map and obtain extinction coefficients for a given set of coordinates. Specifically, there is a handy `dustmaps.csfd.CSFDQuery` method that takes as input an RA, Dec list in SkyCoords format and returns an array of E(B-V) values. An R_V value is used to convert those into extinction coefficients and then used to de-redden SDSS and DES magnitudes. Using SFD maps to obtain E(B-V), then apply band-specific extinction coefficients to get extinction values.
- Subtracting these extinction values $A_{X,CSFD}$ from the observed magnitudes MAG_X to obtain the dereddened magnitudes:

$$MAG_X_CORRECTED = MAG_X - A_X_CSFD$$

For the DES magnitudes, I used the corrected magnitude prescription of Pandey et al. 2021:

$$MAG_CORRECTED = MAG + DELTA_MAG_Y4 + DELTA_MAG_CHROM - A_X_CSFD$$

- (placeholder for pseudocode of algorithm.)
- As one might expect, this had a notable effect on the amplitude of the signal, increasing it by XX%.

3.2.3 Weighted Least-squares fitting.

To fit models to our results and facilitate comparison to the literature, I have added a function (`do_WLS_fit.py`) to perform weighted least-squares fits to extinction profiles. Surprisingly, we hadn't implemented this yet! The function, partially reproduced in Appendix A, is principally a wrapper for the StatsModel package's weighted least-squares method, with the additional capability to rescale weights for fits in logarithmic space (i.e., power laws). This code was used to generate all fits shown in this report, and weights are based on TreeCorr's estimates of the variance at each point in the extinction profile.

¹https://github.com/karllark/dust_extinction

²<https://github.com/greggreen/dustmaps>

4 Catalogs: Round 2

4.1 redMaGiC and WISExSCOS

- I have a note in here about going over magnitude column definitions, but I don't know what that means. Perhaps just the names?
- List counts for randoms and reals
- The WISExSuperCOSMOS catalog is trimmed to $0 < z < 0.2$, rather than $z < 0.11$ as before, as well as $17 < bCalCorr < 21$ (really with that last one?)
- To ensure a decent redshift separation between the background and foreground samples, the high-density redMaGiC sample is restricted to $z > 0.5$ (the full redshift range is $0.15 < z < 0.65?$).
- There is still no WISExSuperCOSMOS random catalog available, so one is created using the corrected randoms algorithm described in Section 3.1 in galactic coordinates to match the original catalog coordinates.

4.2 SDSS and GWSLC

- Detail catalog counts, selections, etc. including magnitude columns.
- Show SDSS query, database
- GWSLC cuts: used X-2 variant of available catalogs, which combines the deepest of the all-sky, medium, and deep surveys. I had originally hoped to examine dust as a function of galaxy type and that is the recommended catalog.
- Don't forget to tabulate wavelengths!

4.3 Systematics testing

- I am always worried that I am screwing up somehow so to test for systematic error I perform my analysis using star catalogs.
- Stars should have no dust so they make good test catalogs either in foreground or in background.
- Randoms: there are strong gradients in density across the fields (makes sense, I guess) which might (seemed to?) affect the results.
- So, to create random catalogs with a matching density gradient, I grabbed larger samples of stars to serve as randoms.

4.3.1 Gaia stars

- This was used as a foreground sample for the redMaGiC catalogs, so it got used for correlations twice: once for the high-density sample, once for the high-luminosity high-redshift sample.
- I had an oddly tough time narrowing down the survey area on Cosmohub, so I ended up grabbing a fairly large area over the Southern sky and applying the redMaGiC mask. Final count: 478,494 objects.

- Detail catalog cuts, which I don't even remember!
- Show one of my pretty overlap plots!

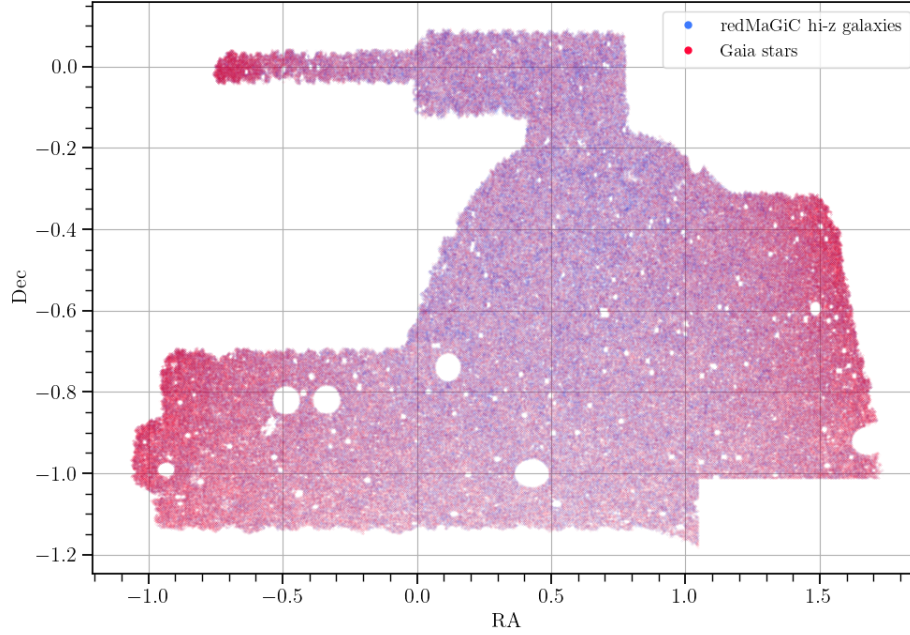


Figure 2: default

4.3.2 DES Y3 stars

- This was used as a background sample, standing in for the redMaGiC catalogs in a correlation with WISExSuperCOSMOS
- Show catalog query used, which was taken from [\[2021ApJS..254...24S\]](#).
- Detail catalog cuts made, which include $18 < \text{mag}_r\text{-corr} < 21$.
- After masking and the like, there were 824,345 stars in the “real” catalog and 9,892,452 randoms.

5 Results

Basic structure of this section: show correlations and best-fit polynomials typeset as an equation. Have my fit summaries from research notes shown verbatim or maybe as a table. The full printout for fits, including R^2 , goes in Appendix.

- Comments for the individual samples are below, but globally, both the SFD and CSFD results agrees with the *slope* of the [\[2010MNRAS.406.1815M\]](#) power law (Equation 13 in that reference) within one sigma. However, the *amplitude* of the CSFD relationship is twice as high as the [\[2010MNRAS.406.1815M\]](#) result. I am not entirely sure what to make of this, but it is what it is.
- Could discuss the SFD results...

5.1 RedMaGiC-based results

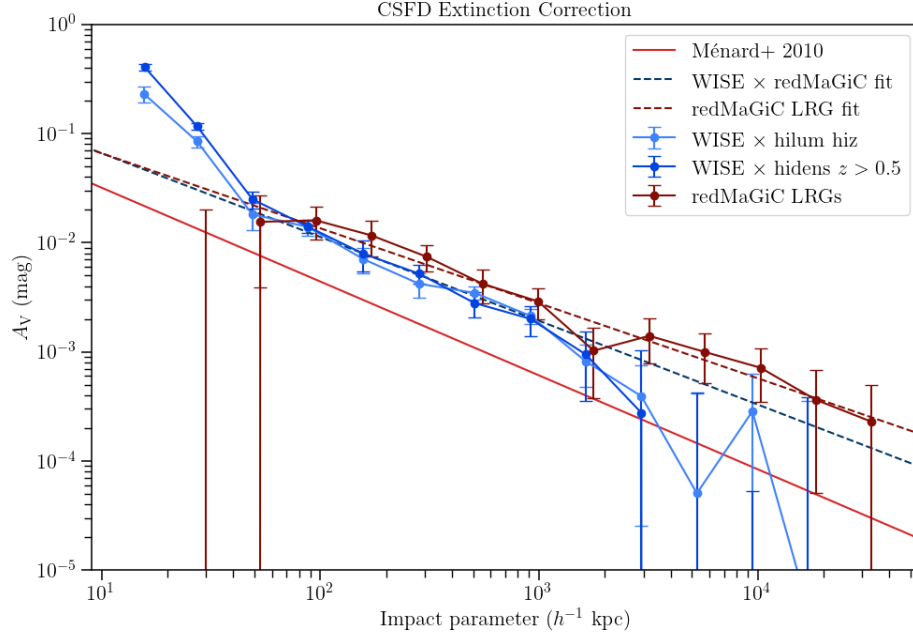


Figure 3: Extinction vs impact parameter for redMaGiC-involved fits. CSFD was used in all cases.

5.1.1 WISExSuperCOSMOS foreground

- Comment on broken power law! May want to compare to literature results
- For one thing, the knee (300 Mpc?) is not quite where the 1- and 2-halo terms traditionally break (1 Mpc). However, that IS a transition Nick has seen in some of his results for the IA position-orientation correlation functions. What that means, idk.

CSFD results:

Hi-dens WLS fit: $1.14\text{e-}02 * (\text{meanr})^{-0.84}$

Hi-z WLS fit: $1.02\text{e-}02 * (\text{meanr})^{-0.72}$

Combined WLS fit: $1.08\text{e-}02 * (\text{meanr})^{-0.77}$

WISExSCOS x redMaGiC high-dens WLS:

Const: $-4.471\text{e+}00 \pm 7.057\text{e-}02$

Slope: $-8.402\text{e-}01 \pm 4.393\text{e-}02$

WISExSCOS x redMaGiC high-lum high-z WLS:

Const: $-4.589\text{e+}00 \pm 1.041\text{e-}01$

Slope: $-7.240\text{e-}01 \pm 5.678\text{e-}02$

Combined WISExSCOS x redMaGiC WLS:

Const: $-4.531\text{e+}00 \pm 6.808\text{e-}02$

Slope: $-7.677\text{e-}01 \pm 3.853\text{e-}02$

5.1.2 redMaGiC LRG

- As expected, there is no power at low impact parameters
- What's fun is that there *is* lots of power at high impact parameter.
- In fact, one COULD argue for the presence of a real 1- to 2-halo transition at about 1 Mpc.

CSFD results:

LRG WLS fit: $8.21\text{e-}03 * (\text{meanr})^{-0.69}$

low-z redMaGiC hi-dens x redMaGiC hi-lum hi-z WLS:

Const: $-4.802\text{e+}00 \pm 2.253\text{e-}01$

Slope: $-6.872\text{e-}01 \pm 5.648\text{e-}02$

5.1.3 Comparison to SFD

Given the acknowledged problems of SFD98, the CSFD formulation of XX is the default used here. It may still be interesting to compare the SFD and CSFD outputs. The difference in extinction profiles is shown in Figure 4. I noticed a couple of interesting trends here. First, there didn't seem to be much of an effect for the high-density sample, but using the SFD extinction correction, the high-lum high-z sample has more extinction at high impact parameters. Put differently, it appears that the SFD prescription added too much extinction at high impact parameters, and with CSFD they are consistent. The interesting/puzzling piece was the redMaGiC LRG

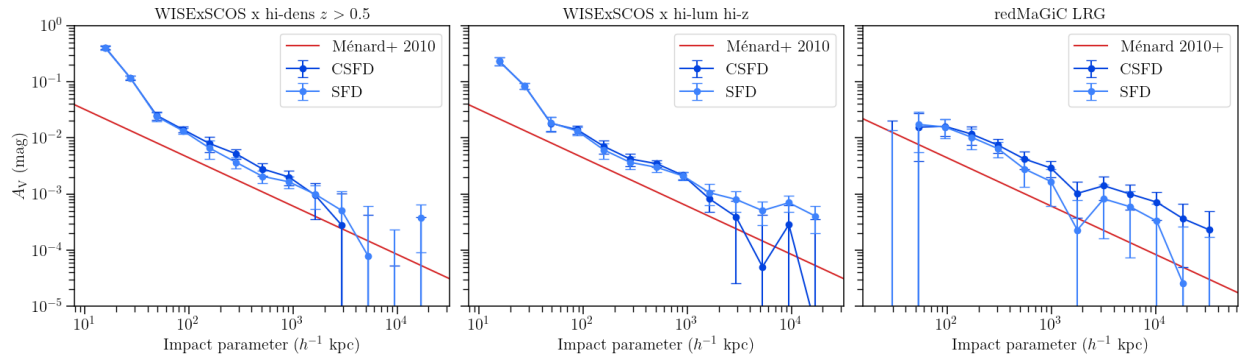


Figure 4: Extinction vs impact parameter for redMaGiC-involved fits. CSFD was used in all cases.

Not CSFD results: had to use a least-squares fit for the LRG, not weighted least squares, since for whatever reason the WLS just didn't want to produce reasonable fits for that sample.

Hi-dens WLS fit: $5.86\text{e-}03 * (\text{meanr})^{-0.62}$

Hi-z WLS fit: $9.51\text{e-}03 * (\text{meanr})^{-0.75}$

Stacked redMaGiC WLS fit: $8.72\text{e-}03 * (\text{meanr})^{-0.74}$

LRG LSQ fit: $6.87\text{e-}03 * (\text{meanr})^{-0.87}$

WISExSCOS x redMaGiC high-dens WLS:

Const: $-5.140\text{e+}00 \pm 3.981\text{e-}01$

Slope: $-6.177\text{e-}01 \pm 1.198\text{e-}01$

WISExSCOS x redMaGiC high-lum high-z WLS:

Const: $-4.656\text{e+}00 \pm 9.686\text{e-}02$

Slope: $-7.525\text{e-}01 \pm 3.224\text{e-}02$

Combined WISExSCOS x redMaGiC WLS:

Const: $-4.742\text{e+}00 \pm 1.444\text{e-}01$

Slope: $-7.392\text{e-}01 \pm 8.036\text{e-}02$

redMaGiC LRG LSQ:

Const: $-4.981\text{e+}00 \pm 3.070\text{e-}01$

Slope: $-8.712\text{e-}01 \pm 1.417\text{e-}01$

5.2 Systematics testing results

This section summarizes the results of the various systematics tests that I implemented using star catalogs. In general, these are mostly consistent with zero, though with noticeable residuals. Then again, these are 10^{-4} results. Note that all of the systematics testing results are based on the CSFD extinction corrections.

5.2.1 Gaia foreground x redMaGiC background

- Show figures
- Show the fit results separately
- Can discuss the fact that high-density sample has a small residual, though I am not entirely sure why

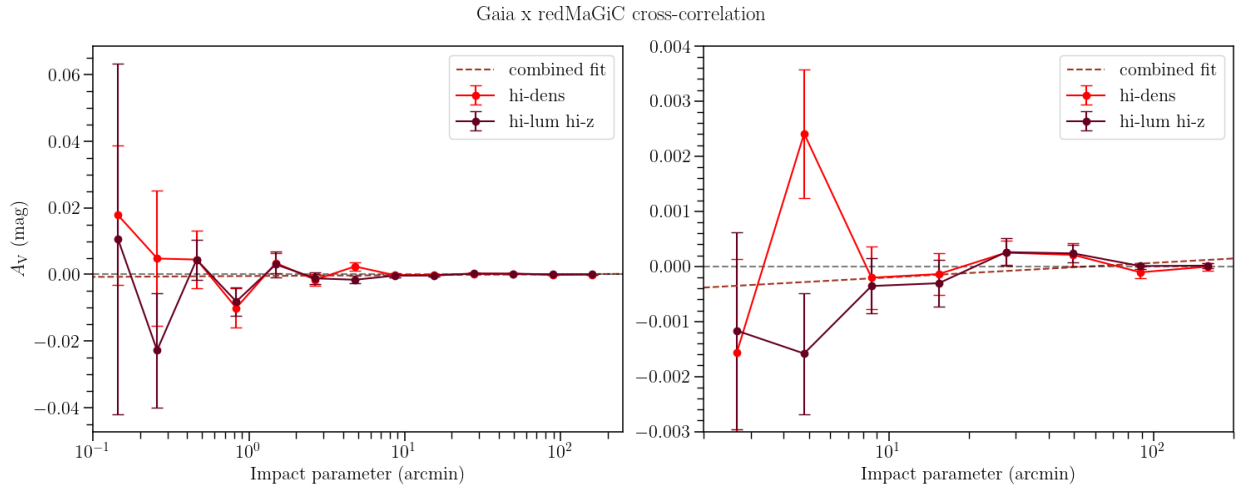


Figure 5: Gaia star foreground catalog cross-correlated with redMaGiC high-luminosity high-redshift (maroon) and high-density (red) background catalogs. The left panel shows the full impact parameter range; the right panel is cropped to show values at higher impact parameters.

Excluding the first two points, which seemed to skew the fits, the results of fitting are

Gaia x redMaGiC high-dens WLS fit:
Const: $3.109\text{e-}04 \pm 4.202\text{e-}04$
Slope: $-6.730\text{e-}05 \pm 8.917\text{e-}05$

Gaia x redMaGiC high-lum high-z WLS fit:
Const: $-3.676\text{e-}05 \pm 3.589\text{e-}04$
Slope: $1.133\text{e-}05 \pm 7.591\text{e-}05$

Gaia x combined redMaGiC WLS fit:
Const: $1.116\text{e-}04 \pm 2.646\text{e-}04$
Slope: $-2.139\text{e-}05 \pm 5.603\text{e-}05$

Gaia x redMaGiC high-dens LSQ fit:
Const: $-4.060\text{e-}04 \pm 1.817\text{e-}03$
Slope: $1.351\text{e-}04 \pm 6.427\text{e-}04$

Gaia x redMaGiC high-lum high-z LSQ fit:
Const: $-5.246\text{e-}04 \pm 1.517\text{e-}03$
Slope: $9.423\text{e-}05 \pm 5.368\text{e-}04$

Gaia x stacked redMaGiC LSQ fit:
Const: $-4.653\text{e-}04 \pm 1.123\text{e-}03$
Slope: $1.146\text{e-}04 \pm 3.974\text{e-}04$

5.2.2 SuperCOSMOSxWISE x DES Y3 stars

- Show figures and fit results for this sample
- Again, small but mildly significant residual.

I decided to go with the least-squares, because the weighted least squares looked like nonsense for this fit. Much bigger slope than made any sense.

WISExSCOS updated rand x DES star WLS fit results:
Coeff: $-5.348\text{e-}04 \pm 4.500\text{e-}04$
Slope: $1.770\text{e-}04 \pm 1.284\text{e-}04$

WISExSCOS updated rand x DES Y3 stars LSQ:
Const: $1.878\text{e-}04 \pm 1.265\text{e-}03$
Slope: $-3.150\text{e-}05 \pm 4.713\text{e-}04$

5.3 SDSS-based results

These results are based on two separate foreground catalogs: the same WISExSuperCOSMOS catalog used for the southern sky, DES-based catalogs, and the GWSLC catalog described above. I will go through these results separately. These are certainly interesting to see because so far we hadn't extended the search to the Northern sky. However, I don't consider the results publication-ready.

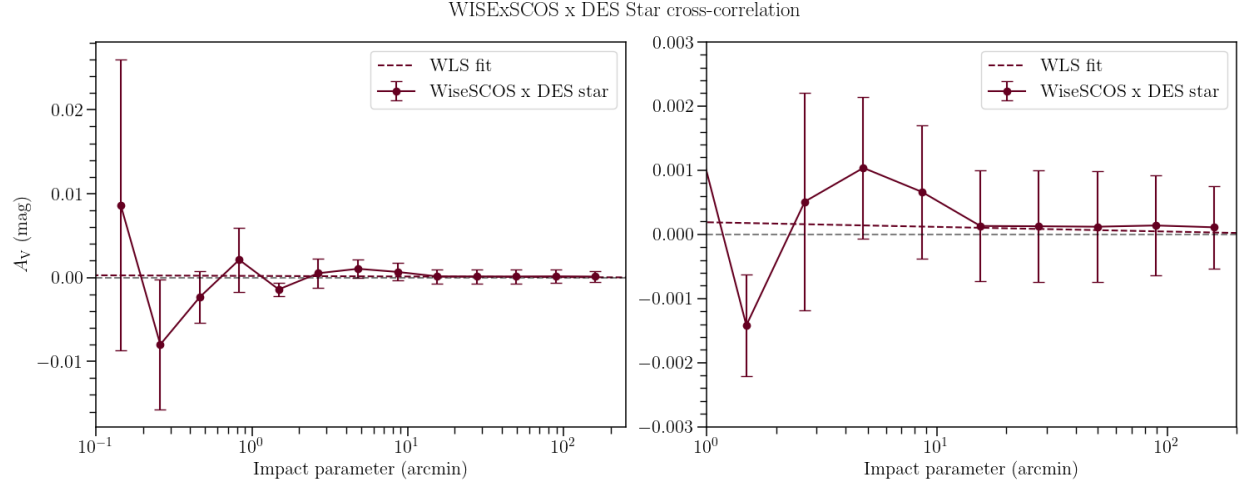


Figure 6: WISExSuperCOSMOS foreground cross-correlated with stars culled from the DES Y3 catalog. The left panel shows the full range of impact parameters considered; the right panel is cropped to show values at higher impact parameters only.

That said, may be an argument to be made for publishing only the *low* impact parameter points. given Eric’s comments about the significance of a detection of a lot of dust at significance, They consistently line up quite well with the redMaGiC-based results at these impact parameters, and Eric did note the significance of the detection of extended dust halos in what is really the extended CGM.

5.3.1 WISExSuperCOSMOS x SDSS

- On the one hand, it’s cool to see this, especially since this is the same foreground sample we use elsewhere that gave us successful results.
- On the other hand, that fat tail is the stickiest result of all my results. Color cuts on the background galaxies, aggressive masking, etc. do nothing. I had even hoped that CSFD, which is supposedly a superior extinction correction algorithm, would make it go away. No dice!
- Since what characterizes the redMaGiC sample is tightness in color space, I also tried to use the eBOSS LRG sample (maybe create one of my own?) but the source density was too low to be used.

5.3.2 GSWLC x SDSS

- This DOES look cool, particularly the fact that at higher impact parameters, the slope lines up very nicely with the redMaGiC samples.
- I did some testing with sSFR [show my algorithm?] but results were (at the time) mysterious to me. the selection functions are also quite heterogenous between quiescent and star-forming samples. Not sure how significant that latter fact is, but it’s there.
- Important to mention: they all seem to depend heavily on the NSIDE of the mask I end up creating. This is a possible indication of my mask being incomplete or wrong; in any case

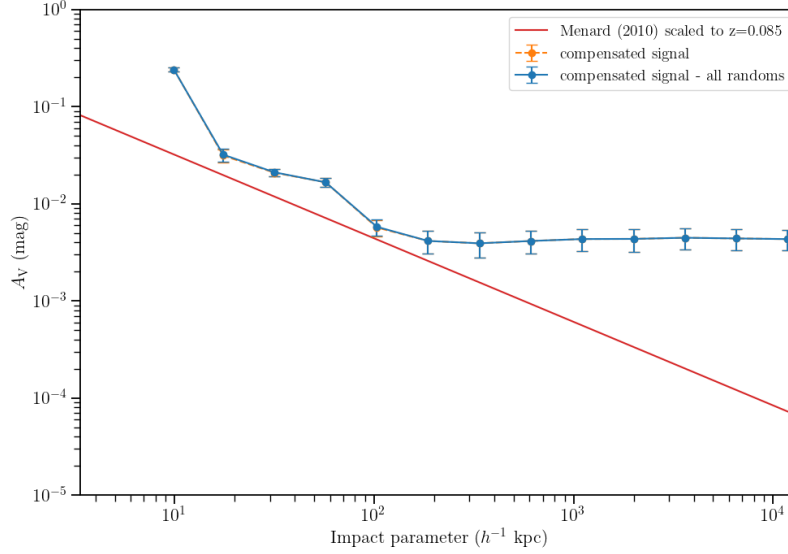


Figure 7: Extinction curve of WISExSuperCOSMOS foreground cross-correlated with SDSS background galaxies, shown as blue points. The M2010 prediction is shown in red.

this is why I don't 100% trust the result

- Plots to show: the neat spatial overlap plots, as well as the ugly multicolor plot which does a good job of demonstrating relationship between the various results, including this one.

Spatial overlap plot: put this into catalogs.

6 Discussion and Next Steps

6.1 Comment on results

- Fact of the matter is that I have been playing with this for years and the results have been consistent.
- The long time period has afforded me the opportunity to modernize, like 2023 dust relation, CSFD, WLS, etc.
- The fact that CSFD is high is consistent with the claim of [2023ApJ...958..118C] that “when SFD is used to derive absolute photometry for sources, Galactic extinction will be overcorrected in a redshift- and spatially-dependent manner following the pattern of the cosmic web.” Now, which is more believable is at the reader's discretion!
- Also, to some extent, it is to be expected that amplitude of signal is proportional to volume covered in foreground (right?). Then again, volume of Ménard sample is significantly higher than ours. As far as I know, that study has no selection for red sequence galaxies [Not sure what I meant by this – ed.]
- To end on a bright note, we know that the redMaGiC galaxies are biased tracers of LSS (by biased here I mean have a high galaxy bias relative to some field galaxy). The presence of dust well into the two-halo regime, while the WISExSuperCOSMOS samples (which presumably

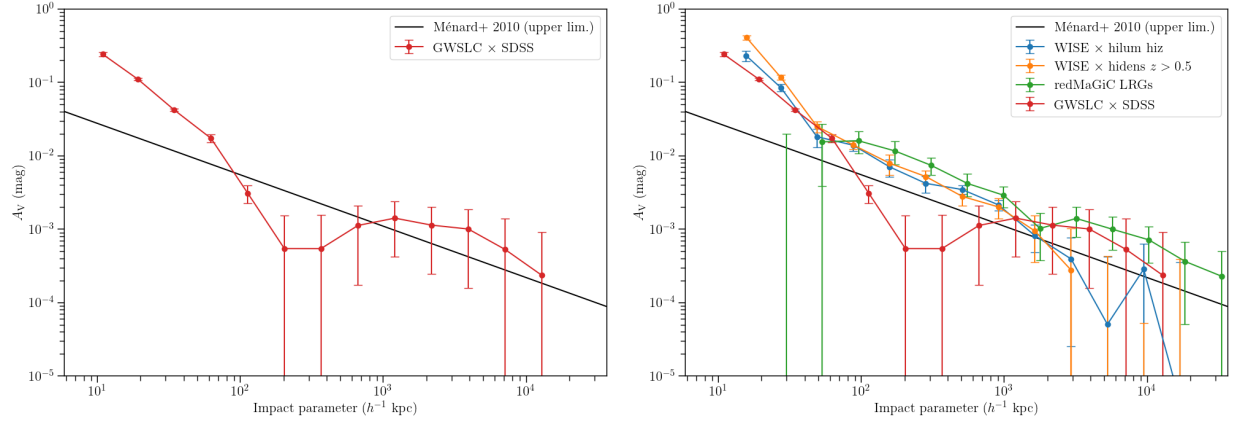


Figure 8: Extinction curve resulting from cross-correlation of GSWLC foreground and SDSS background galaxies. The GSWLC-SDSS result is shown in isolation in the left panel in red and together with the redMaGiC-based results in the right panel: WISExSuperCOSMOS–high-luminosity high-redshift (blue), WISExSuperCOSMOS–high density (orange), and the redMaGiC LRG fit (green). The M2010 prediction is shown in black in both panels.

include star-forming galaxies) fall off, tells a cool story about redMaGiC as a tracer of LSS and WISExSuperCOSMOS a little more general.

6.2 Next steps

- At this point, I would like to publish. One can fiddle with results ad infinitum, but I’ve been urged by multiple people to publish so I will. If you all want to start running systematic analyses, you can, but I am not going to wait around.
- I figure this is a beginning, not an end. In follow-on studies, we can do calibration, segregate by galaxy type, etc. (Note that I had tried to do a galaxy type study (motivation for SDSS/GWSLC!) but my results were inconclusive and I haven’t redone them since.)
- For northern-sky samples, it might be worth trying Pan-STARRS instead of SDSS, or perhaps KiDS or some other such thing. I don’t know what the benefits would be of a more “modern” survey over SDSS, but it would be interesting to see.
- Jim, still waiting on your emission study! Also, Eric, could we transfer ownership of main repository to me? Alternatively, I can sync my fork.
- My ultimate goal is to make a function or tool that can help calibrate SN extinction along the line of sight as a function of impact parameter.

A Appendix: Weighted least-squares fit and results

The important parts of `do_WLS_fit.py` are reproduced below.

```
import statsmodels.api as sm
import numpy as np
```

```
def do_WLS_fit(this_dk, dust_rel, name=None, start=None, end=None, log=False):
```

```

"""
Perform a weighted least-squares linear regression to
data, taking care to transform the variables appropriately.

Inputs
    this_dk: should be the 'compensated' treecorr output
    dust_rel: the corrected A_V relationship (dk-dr+rr)
    name: what are we calling it?
    start: exclude points with index < 'start' for fitting
    end: ibidem, but for points with index > 'end'
    log: if True, fit the relationship in log-space to stabilize the fits.
        Default is False.
"""

## If you want to exclude some points at the start/end
dust_rel = dust_rel[start:end]
this_dk = this_dk[start:end]

## If doing a logarithmic fit, transform the
## dependent variable (the fully compensated TreeCorr kappa)
if log == True:
    # Keep positive values only
    ok = (dust_rel > 0)
    dust_rel = dust_rel[ok]
    this_dk = this_dk[ok]

    # Fitting log of dust relation, so take log
    Y = np.log(dust_rel)

    ## Also need to transform the weights!
    weights_0G = 1/(this_dk['sigma']**2)
    #weight_scale = (1/Y)**2
    weight_scale = (1/(Y+1))**2
    weights = weights_0G * weight_scale

## If not doing a logarithmic fit, don't transform
else:
    Y = dust_rel
    weights = 1/(this_dk['sigma']**2)

## If plotting X on a log-scale, which we are,
## fit relationship on log scale
X = this_dk['meanlogr']

## Add intercept to abscissa
X = sm.add_constant(X)

## Do fit

```

```

mod_wls = sm.WLS(Y, X, weights=weights)
res_wls = mod_wls.fit()

## Print results
print("")
print(f"Fit result: {name}")
print("")
print(res_wls.summary())

return(res_wls)

```

Appendix: SDSS queries

Does this actually need to be here? Does anyone care?

B Full WLS Regression results

Full printouts of the `statsmodels.api.WLS` regressions are reproduced below. For real fits (not the systematics tests), the constant term is converted to a coefficient as e^{const} in the power law.

B.1 WISExSCOS and redMaGiC fits

The fit results below are based on only the “flat” portions of the extinction curves in Figure 3, i.e., 50-1000 kpc for the CSFD fits, and results are based on a high-redshift subset ($z_{\text{redmagic}} > 0.5$) of the redMaGiC high-density catalog.

WLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.975			
Model:	WLS	Adj. R-squared:	0.973			
Method:	Least Squares	F-statistic:	397.0			
Date:	Wed, 23 Oct 2024	Prob (F-statistic):	2.23e-09			
Time:	18:59:41	Log-Likelihood:	8.2589			
No. Observations:	12	AIC:	-12.52			
Df Residuals:	10	BIC:	-11.55			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-4.5308	0.068	-66.556	0.000	-4.683	-4.379
x1	-0.7677	0.039	-19.925	0.000	-0.853	-0.682
=====						
Omnibus:	1.628	Durbin-Watson:	2.041			
Prob(Omnibus):	0.443	Jarque-Bera (JB):	1.154			
Skew:	-0.692	Prob(JB):	0.562			
Kurtosis:	2.373	Cond. No.	6.06			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[-4.53082712 -0.7676558]

Fit: $1.08e-02 * (\text{meanr})^{-0.77}$

Below is the full fit result for the redMaGiC LRG sample, i.e., the low-redshift subset ($z_{\text{redmagic}} > 0.5$) of the redMaGiC high-density sample cross-correlated with the high-luminosity high-redshift

Fit result: redmagic LRG

WLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.943			
Model:	WLS	Adj. R-squared:	0.936			
Method:	Least Squares	F-statistic:	148.0			
Date:	Wed, 23 Oct 2024	Prob (F-statistic):	6.85e-07			
Time:	18:59:41	Log-Likelihood:	-2.4161			
No. Observations:	11	AIC:	8.832			
Df Residuals:	9	BIC:	9.628			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-4.8020	0.225	-21.318	0.000	-5.312	-4.292
x1	-0.6872	0.056	-12.166	0.000	-0.815	-0.559
=====						
Omnibus:	7.605	Durbin-Watson:	1.874			
Prob(Omnibus):	0.022	Jarque-Bera (JB):	3.122			
Skew:	-1.072	Prob(JB):	0.210			
Kurtosis:	4.487	Cond. No.	13.8			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[-4.80200583 -0.68720111]

Fit: $8.21e-03 * (\text{meanr})^{-0.69}$

B.2 Gaia x redMaGiC

Gaia foreground cross-correlated with redMaGiC high-luminosity high-redshift sample.

Fit result: Gaia x hi-lum hi-z

WLS Regression Results

```

=====
Dep. Variable:          y      R-squared:          0.002
Model:                  WLS    Adj. R-squared:      -0.089
Method:                  Least Squares  F-statistic:      0.02226
Date:                    Wed, 23 Oct 2024  Prob (F-statistic):      0.884
Time:                    18:48:14  Log-Likelihood:      70.136
No. Observations:        13      AIC:              -136.3
Df Residuals:            11      BIC:              -135.1
Df Model:                 1
Covariance Type:         nonrobust
=====

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const      -3.676e-05      0.000      -0.102      0.920      -0.001      0.001
x1          1.133e-05      7.59e-05      0.149      0.884      -0.000      0.000
=====

```

```

=====
Omnibus:          0.812  Durbin-Watson:          2.270
Prob(Omnibus):    0.666  Jarque-Bera (JB):          0.628
Skew:             0.015  Prob(JB):          0.730
Kurtosis:         1.923  Cond. No.          51.2
=====

```

Gaia foreground cross-correlated with redMaGiC high-density sample.

Fit result: Gaia x high-dens

WLS Regression Results

```

=====
Dep. Variable:          y      R-squared:          0.049
Model:                  WLS    Adj. R-squared:      -0.037
Method:                  Least Squares  F-statistic:      0.5696
Date:                    Wed, 23 Oct 2024  Prob (F-statistic):      0.466
Time:                    18:48:14  Log-Likelihood:      69.030
No. Observations:        13      AIC:              -134.1
Df Residuals:            11      BIC:              -132.9
Df Model:                 1
Covariance Type:         nonrobust
=====

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const          0.0003      0.000      0.740      0.475      -0.001      0.001
x1            -6.73e-05      8.92e-05     -0.755      0.466      -0.000      0.000
=====

```

```

=====
Omnibus:          0.378  Durbin-Watson:          3.063
Prob(Omnibus):    0.828  Jarque-Bera (JB):          0.490
Skew:            -0.175  Prob(JB):          0.783
Kurtosis:         2.115  Cond. No.          36.3
=====

```


Finally, Gaia foreground cross-correlated with the combined redMaGiC galaxy sample.

Fit result: Gaia x combined redMaGiC

WLS Regression Results						
=====						
Dep. Variable:	y	R-squared:	0.006			
Model:	WLS	Adj. R-squared:	-0.035			
Method:	Least Squares	F-statistic:	0.1458			
Date:	Wed, 02 Oct 2024	Prob (F-statistic):	0.706			
Time:	18:24:35	Log-Likelihood:	138.85			
No. Observations:	26	AIC:	-273.7			
Df Residuals:	24	BIC:	-271.2			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.0001	0.000	0.422	0.677	-0.000	0.001
x1	-2.139e-05	5.6e-05	-0.382	0.706	-0.000	9.42e-05
=====						
Omnibus:	1.496	Durbin-Watson:	2.023			
Prob(Omnibus):	0.473	Jarque-Bera (JB):	0.966			
Skew:	-0.059	Prob(JB):	0.617			
Kurtosis:	2.063	Cond. No.	45.5			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[1.11602366e-04 -2.13928514e-05]

Gaia x combined redMaGiC WLS fit:

Const: 1.116e-04 +/- 2.646e-04

Slope: -2.139e-05 +/- 5.603e-05

B.3 WISExSuperCOSMOS x DES stars

Standard WISExSuperCOSMOS foreground sample cross-correlated with DES Y3 star sample background. The rand2 refers to the new random-catalog generation algorithm I used, which

Fit result: WISExSCOS rand2 x DES star

WLS Regression Results			
Dep. Variable:	y	R-squared:	0.147
Model:	WLS	Adj. R-squared:	0.070
Method:	Least Squares	F-statistic:	1.900

Date: Wed, 02 Oct 2024 Prob (F-statistic): 0.195
Time: 18:24:36 Log-Likelihood: 69.816
No. Observations: 13 AIC: -135.6
Df Residuals: 11 BIC: -134.5
Df Model: 1
Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
const	-0.0005	0.000	-1.188	0.260	-0.002	0.000
x1	0.0002	0.000	1.378	0.195	-0.000	0.000
Omnibus:		0.407	Durbin-Watson:			1.947
Prob(Omnibus):		0.816	Jarque-Bera (JB):			0.509
Skew:		-0.268	Prob(JB):			0.775
Kurtosis:		2.193	Cond. No.			7.93

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[-0.00053476 0.000177]

WISExSCOS updated rand x DES star WLS fit results:

Coeff: -5.348e-04 +/- 4.500e-04

Slope: 1.770e-04 +/- 1.284e-04

B.4 GWSLC x SDSS

This is the fit result for a subset of the GWSLC X-2 foreground catalog cross-correlated with a background catalog selected from SDSS. In this case,

Fit result: partial GWSLC x SDSS

WLS Regression Results						
Dep. Variable:	y	R-squared:				0.969
Model:	WLS	Adj. R-squared:				0.961
Method:	Least Squares	F-statistic:				124.3
Date:	Tue, 15 Oct 2024	Prob (F-statistic):				0.000368
Time:	09:08:29	Log-Likelihood:				0.91149
No. Observations:	6	AIC:				2.177
Df Residuals:	4	BIC:				1.761
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-4.2028	0.264	-15.926	0.000	-4.935	-3.470

x1	-0.7612	0.068	-11.148	0.000	-0.951	-0.572
=====						
Omnibus:		nan	Durbin-Watson:			1.677
Prob(Omnibus):		nan	Jarque-Bera (JB):			0.340
Skew:		0.319	Prob(JB):			0.844
Kurtosis:		2.025	Cond. No.			10.5
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
 [-4.20276501 -0.7611631]

GWSLC x SDSS Fit: 1.45e-02 * (meanr)^-0.75
 partial GWSLC x SDSS Fit: 1.50e-02 * (meanr)^-0.76

References

- [1] Karl D. Gordon. “dust_extinction: Interstellar Dust Extinction Models”. In: *Journal of Open Source Software* 9.100 (2024), p. 7023. DOI: 10.21105/joss.07023. URL: <https://doi.org/10.21105/joss.07023>.
- [2] Karl D. Gordon et al. “One Relation for All Wavelengths: The Far-ultraviolet to Mid-infrared Milky Way Spectroscopic R(V)-dependent Dust Extinction Relationship”. In: *Astrophysical Journal* 950.2, 86 (June 2023), p. 86. DOI: 10.3847/1538-4357/accb59. arXiv: 2304.01991 [astro-ph.GA].